COMP 421 - Database Systems

Project 3: Application

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1 Relational Model

The relational model has not been modified from Project 2.

Model

`HealthcareInstitution(hcId, phoneNum, name, email, address, website)`

`CommunityClinic(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

`BirthingCenter(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

`Midwife(pracId, name, email, phoneNum, hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

note: `hcId` has to be `NOT NULL` to enforce Midwife's participation constraint in the `employs` relationship set.

`Father(fatherId, phoneNum, name, email, address, QChcardid, dateOfBirth, profession, bloodType)`

`Mother(QChcardid, phoneNum, name, email, adddress, dateOfBirth, profession, bloodType)`

`Couple(coupleId, invited, interested, selected, QChcardid, fatherId)`

`QChcardid` foreign key referencing `Mother`
`fatherId` foreign key referencing `Father`

note: `QChcardid` has to be `NOT NULL` to enforce Couple's participation constraint in the `constitutesF` relationship set.

`InformationSession(sessId, language, date, time, pracId)`

`pracId` foreign key referencing Midwife

note: `pracId` has to be `NOT NULL` to enforce InformationSession's participation constraint in the `hosts` relationship set.

`Invitation(sessId, coupleId, attended, registered)`

`sessId` foreign key referencing `Information Session`
`coupleId` foreign key referencing `Couple`

`Pregnancy(coupleId, pregNum, estDueDate, expBirthTimeframe, lastMenstPeriod, isHomeBirth, primaryMidwife, backupMidwife)`

`coupleId` foreign key referencing `Couple`
`primaryMidwife` foreign key referencing `Midwife(pracId)`
`backupMidwife` foreign key referencing `Midwife(pracId)`

note: `coupleId` is part of the primary key and thus has to be `NOT NULL`.

Appointment(apptId, date, time, pregNum, coupleId, pracId)

(pregNum, coupleId) foreign key referencing Pregnancy
pracId foreign key referencing Midwife

note: pregNum and coupleId has to be NOT NULL to enforce Appointment's participation constraint in the belongs relationship set.

note: pracId has to be NOT NULL to enforce Appointment's participation constraint in the sets up relationship set.

Note(noteId, timestamp, description, apptId)

apptId foreign key referencing Appointment

note: apptId has to be NOT NULL to enforce Note's participation constraint in the belongs relationship set.

Baby(babyId, name, gender, timeOfBirth, dateOfBirth, bloodType, pregNum, coupleId)

(pregNum, coupleId) foreign key referencing Pregnancy

note: pregNum and coupleId has to be NOT NULL to enforce Baby's participation constraint in the born from relationship set.

Technician(techId, name, phoneNum)

Test(testId, type, datePrescribed, dateSampleTaken, results, dateConducted, lab, techId, pracId, pregNum, coupleId, babyId, QChcardid)

techId foreign key referencing Technician
pracId foreign key referencing Midwife
(pregNum, coupleId) foreign key referencing Pregnancy
babyId foreign key referencing Baby
QChcardid foreign key referencing Mother

note: pregNum, coupleId has to be NOT NULL to enforce Test's participation constraint in the belongs relationship set.

note: techId has to be NOT NULL to enforce Test's participation constraint in the conducts relationship set.

note: pracId has to be NOT NULL to enforce Test's participation constraint in the prescribes relationship set.

BirthLocation(hcId, pregNum, coupleId) hcId foreign key referencing BirthingCenter

(pregNum, coupleId) foreign key referencing Pregnancy

Note: relation names have been pluralized in the db tables to follow convention.

Artificial keys

Entity Set	Artificial key
HealthcareInstitution	hcId
Father	fatherId
InformationSession	sessId
Test	testId
Baby	babyId
Appointment	apptId
Note	noteId
Couple	coupleId

Table 1: Artificial keys created in the relational model

2 Application Interaction

a. **Assume: `P0` = 123.**

i Run a query in the database that shows that midwife with practitioner id `P0` does not exist.

```
db2 => SELECT name FROM Midwives WHERE pracId = 123;

NAME
------------------------------

  0 record(s) selected.
```

Figure 1: a (i)

ii Start the application and enter the practitioner id (of the midwife) `P0`. **ii.**

```
szinat@winter2022-comp421:~$ java GoBabbyApp
----- Go Babby Application ------

Please enter your practitioner id [E] to exit: 123
Invalid practitioner id

Please enter your practitioner id [E] to exit:
```

Figure 2: a (ii)

b. **Assume:**

- **mother `M1` is named *Victoria Gutierrez***
- **mother `M2` is named *Corina Solis***
- **Appointment date `D1` = 2022/03/21.**
- **Note `N1`'s observation is *This is my newest note.***

i Run a query in the database that shows that a mother M1 has had at least two pregnancies (can be as part of the same or different couple - does not matter).

```
db2 => SELECT C.COUPLEID, PREGNUM, NAME, C.QCHCARDID, PRIMARYMIDWIFE, BACKUPMIDWIFE FROM PREGNANCIES
JOIN COUPLES C on PREGNANCIES.COUPLEID = C.COUPLEID
JOIN MOTHERS M on C.QCHCARDID = M.QCHCARDID
WHERE NAME='Victoria Gutierrez';
db2 (cont.) => db2 (cont.) => db2 (cont.) =>
COUPLEID    PREGNUM     NAME                           QCHCARDID    PRIMARYMIDWIFE BACKUPMIDWIFE
----------- ----------- ------------------------------ ------------ -------------- -------------
       4010           1 Victoria Gutierrez             VIGU63128923           1005          1002
       4010           2 Victoria Gutierrez             VIGU63128923           1010          1001
       4010           3 Victoria Gutierrez             VIGU63128923           1001          1002

  3 record(s) selected.
```

Figure 3: b (i)

ii In the application, enter the practitioner id (of the midwife) `P1` (must exist in the database). Assume `P1` = 1001.

```
szinat@winter2022-comp421:~$ java GoBabbyApp
----- Go Babby Application ------

Please enter your practitioner id [E] to exit: 1001
Welcome Marion

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit:
```

Figure 4: b (ii)

iii Enter the date `D1`. One of the appointments must be for `M1` and another one for `M2` , a different expecting mother.

```
szinat@winter2022-comp421:~$ java GoBabbyApp
----- Go Babby Application ------

Please enter your practitioner id [E] to exit: 1001
Welcome Marion

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit: 2022-03-21
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
       [E] to exit [D] to go back to another date:
```

Figure 5: b (iii)

iv Choose the appointment number associated with `M1`.

```
Please enter the date (YYYY-MM-DD) for appointment list [E] to exit: 2022-03-21
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: 1

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 6: b (iv)

v Enter 1. (Should show at least two notes).

```
Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: 1

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 1
2022-02-13 12:23:32.0   Couple would prefer home birth.
2022-02-13 10:02:45.26  Baby has good movements

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments
```

Figure 7: b (v)

vi Enter 2. (Should show at least two tests).

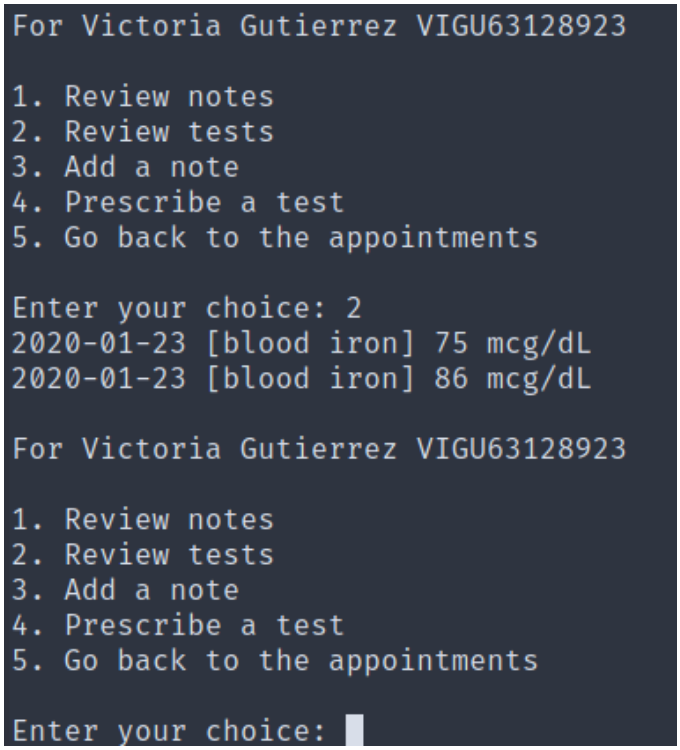


Figure 8: b (vi)

vii Enter 3. Add a note `N1`.

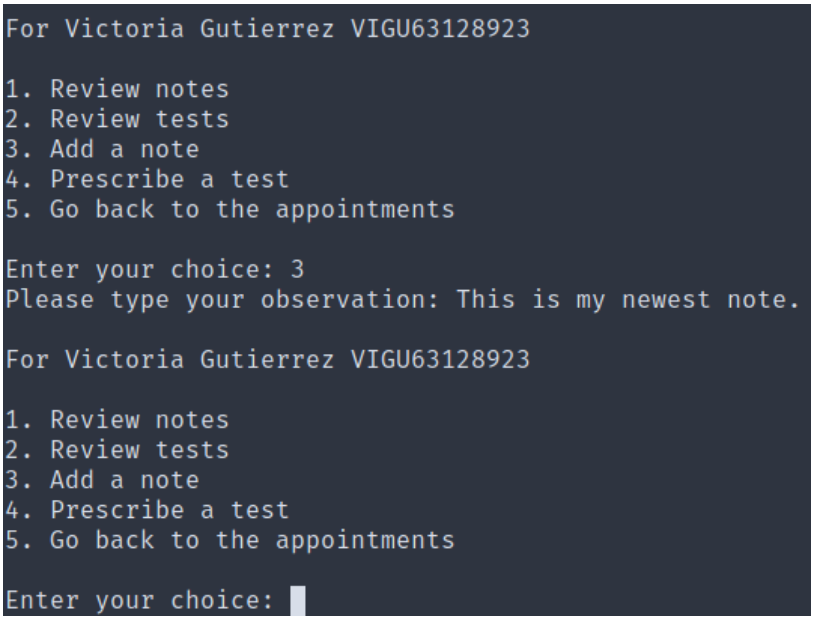


Figure 9: b (vii)

viii Run a query in the database that shows all the notes associated with M1 , include couple information, pregnancy number, etc., as relevant in your model. The actual note text may be truncated to first 50 characters or so to reduce clutter.

```
db2 => SELECT A.COUPLEID, A.PREGNUM, NAME, C.QCHCARDID, PRACID, CAST (DESCRIPTION AS VARCHAR(35))
FROM NOTES
         JOIN APPOINTMENTS A on NOTES.APPTID = A.APPTID
         JOIN PREGNANCIES P on A.PREGNUM = P.PREGNUM and A.COUPLEID = P.COUPLEID
         JOIN COUPLES C on P.COUPLEID = C.COUPLEID
         JOIN MOTHERS M on C.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Victoria Gutierrez';db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>

COUPLEID    PREGNUM     NAME                           QCHCARDID    PRACID      6
----------- ----------- ------------------------------ ------------ ----------- ---------------------------------
       4010           2 Victoria Gutierrez             VIGU63128923        1001 Couple would prefer home birth.
       4010           2 Victoria Gutierrez             VIGU63128923        1001 This is my newest note.
       4010           2 Victoria Gutierrez             VIGU63128923        1001 Baby has good movements

  3 record(s) selected.
```

Figure 10: b (viii)

ix Enter 1.

```
For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 1
2022-03-13 11:15:04.742 This is my newest note.
2022-02-13 12:23:32.0   Couple would prefer home birth.
2022-02-13 10:02:45.26  Baby has good movements

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 11: b (viii)

x Enter 4. Add a test for trisomy 21.

```
For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 4
Please enter the type of test: trisomy 21

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 12: b (x)

xi Run a query in the database that shows all the tests associated with `M1` , include couple information, pregnancy number, etc., as relevant in your model. The result attribute of the test maybe truncated to the first 50 characters or so.

```
db2 => SELECT TYPE, NAME, M.QCHCARDID, COUPLEID, PREGNUM, CAST(RESULTS AS VARCHAR(35))
FROM TESTS
        JOIN MOTHERS M on TESTS.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Victoria Gutierrez';db2 (cont.) => db2 (cont.) => db2 (cont.) =>

TYPE                           NAME                           QCHCARDID    COUPLEID    PREGNUM     6
------------------------------ ------------------------------ ------------ ----------- ----------- ------------
trisomy 21                     Victoria Gutierrez             VIGU63128923        4010           2 -
blood iron                     Victoria Gutierrez             VIGU63128923        4010           2 75 mcg/dL
blood iron                     Victoria Gutierrez             VIGU63128923        4010           2 86 mcg/dL

  3 record(s) selected.
```

Figure 13: b (xi)

xii Enter 5 (see Figure 15 - c(ii))

c. **Assume: mother `M2` is named *Corina Solis***

i Run a query in the database that shows all the tests associated with `M2` , include couple information, pregnancy number, etc., as relevant in your model. The result attribute of the test maybe truncated to the first 50 characters or so.

```
db2 => SELECT TYPE, NAME, M.QCHCARDID, COUPLEID, PREGNUM, CAST(RESULTS AS VARCHAR(35)) AS RESULTS
FROM TESTS
        JOIN MOTHERS M on TESTS.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Corina Solis';
db2 (cont.) => db2 (cont.) => db2 (cont.) =>
TYPE                          NAME                          QCHCARDID    COUPLEID    PREGNUM     RESULTS
----------------------------- ----------------------------- ------------ ----------- ----------- -------------
blood iron test               Corina Solis                  CASL23451234        4002           1 -
sonography                    Corina Solis                  CASL23451234        4002           1 -

  2 record(s) selected.
```

Figure 14: c (i)

ii Choose the appointment associated with mother `M2`

```
For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 5
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: 2

For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 15: c (ii)

iii Enter 1 (to see the relevant notes).

```
For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 1
2022-03-13 09:24:05.26  Everything is going well yay!
2022-03-12 20:07:17.658 this is great!

For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 16: c (iii)

iv Enter 4. Add a test prescription for Group B streptococcus (test type).

```
For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 4
Please enter the type of test: Group B streptococcus

For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 17: c (iv)

v Run a query in the database that shows all the tests associated with `M2` , include couple information, pregnancy number, etc., as relevant in your model. The result attribute of the test maybe truncated to the first 50 characters or so.

```
db2 => SELECT TYPE, NAME, M.QCHCARDID, COUPLEID, PREGNUM, CAST(RESULTS AS VARCHAR(35)) AS RESULTS
FROM TESTS
        JOIN MOTHERS M on TESTS.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Corina Solis';
db2 (cont.) => db2 (cont.) => db2 (cont.) =>
TYPE                           NAME                           QCHCARDID    COUPLEID    PREGNUM     RESULTS
------------------------------ ------------------------------ ------------ ----------- ----------- --------
Group B streptococcus          Corina Solis                   CASL23451234        4002           1 -
blood iron test                Corina Solis                   CASL23451234        4002           1 -
sonography                     Corina Solis                   CASL23451234        4002           1 -

  3 record(s) selected.
```

Figure 18: c (v)

d. **Assume appointment date `D2` = 2022/05/29**

i Choose 5, followed by D.

```
For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 5
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: d

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit:
```

Figure 19: d (i)

ii Enter a different date D2

```
Enter your choice: 5
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: d

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit: 2022-05-29
1:      07:45:00 B Judith Leon JULE09156234
2:      08:45:00 B Bobbi Cherry BOCH07562649

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date:
```

Figure 20: d (ii) and d (iii)

iii It should show at least two appointments for mothers M3 , M4 , who are not the same as previous examples.

3 Indexing

Create an index that will help retrieve a mother's address, assuming that we only have a phone number to start the search with.

```
db2 => CREATE UNIQUE INDEX idx1 -- the phoneNum has to be unique for indexing to be meaningful here
    ON MOTHERS (PHONENUM) INCLUDE (ADDRESS);db2 (cont.) =>
DB20000I  The SQL command completed successfully.
db2 =>
```

Figure 21: Q4

4 Data Analytics

```
db2 => EXPORT TO "/home/2022/szinat/birth_months.csv" OF DEL
SELECT *
FROM (
        SELECT SUBSTRING('JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC ', monthNo * 4 - 3, 3) AS birthMonth,
               numBirth
        FROM (
                SELECT MONTH(DATEOFBIRTH) AS monthNo, COUNT(DATEOFBIRTH) AS numBirth
                FROM BABIES
                WHERE DATEOFBIRTH IS NOT NULL
                GROUP BY MONTH(DATEOFBIRTH)
                ORDER BY MONTH(DATEOFBIRTH)
            )
);db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2
 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>
SQL3104N  The Export utility is beginning to export data to file
"/home/2022/szinat/birth_months.csv".

SQL3105N  The Export utility has finished exporting "12" rows.


Number of rows exported: 12
```

Figure 22: SQL export to export db query into csv file

Plot generated using matplotlib. submitted python code: `data_analytics.py`.

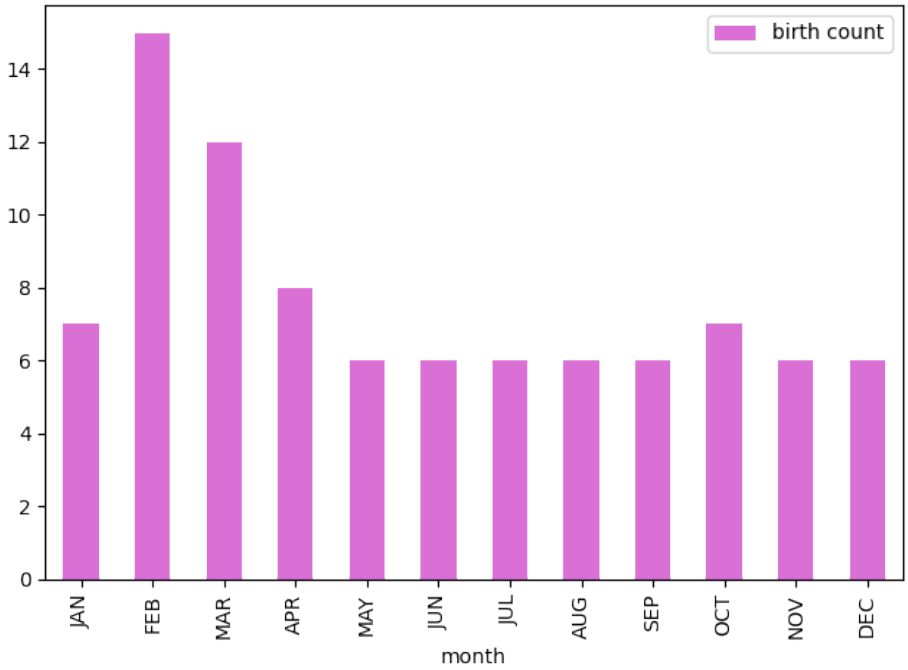


Figure 23: Bar plot showing the number of births in each month